Robust Contextual State Estimation with Limited Measurement Data

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***Abstract*—The increasing integration of distributed energy resources (DERs) is transforming power systems into complex, decentralized networks, particularly at the distribution level, where active distribution networks (ADNs) introduce new challenges for monitoring and control. Accurate state estimation in these systems is complicated by limited and heterogeneous measurement infrastructures, where real-time data from supervisory control and data acquisition (SCADA) and microphasor measurement units (μ-PMUs) is sparse, and slower data from Advanced Metering Infrastructure (AMI) is abundant but delayed. In this context, historical data becomes a key asset, yet its effective use requires robust methods that account for its relevance and variability. This paper proposes a novel data-driven methodology for state estimation under limited real-time observability. We first introduce a non-parametric estimator that leverages nearest neighbors to approximate the conditional expectation of delayed measurement residuals. Building on this, we develop a robustified extension that accounts for ambiguity in the estimated conditional distribution, mitigating the impact of sample sparsity and uncertainty. Numerical experiments demonstrate that the proposed approach brings substantial benefits in accuracy and robustness, particularly in scenarios with scarce and noisy measurements.**

***Index Terms*—Real-time observability, active distribution networks, distribution system state estimation, robust estimation, contextual stochastic optimization**

I. Introduction

POWER systems are undergoing a profound transformation, shifting from traditionally centralized architectures toward increasingly complex and decentralized networks. This evolution is driven by the growing penetration of DERs, such as rooftop solar, energy storage, and electric vehicles, which introduce significant operational uncertainty due to their stochastic behavior both in the generation and in the consumption. At the distribution level in particular, ADNs characterized by bidirectional power flows, tighter interconnections and changing topology, emerge as a new paradigm, making their operation and monitoring substantially more challenging than in the past [1].

Accurate state estimation in such uncertain environments remains a critical task for real-time monitoring and control. However, measurement infrastructure in distribution systems is typically sparse and heterogeneous [2]: fast but sparse measurements like SCADA systems and μ-PMUs are received in real-time but exist in limited number, while slow but abundant measurements consist of more widely available data from AMI such as Smart Meters (SM). This limitation in granularity and latency leads most existing approaches to favor static estimation frameworks over dynamic and forecasting-aided ones [3].

In the face of limited real-time measurements, historical data becomes a valuable resource for inferring the current state of the system. However, in modern distribution networks, even assuming that the grid topology and parameters are known, we typically face a *double information shortage*. First, the available real-time measurements are often insufficient to guarantee system observability, especially given the sparse and uneven deployment of sensing infrastructure. Second, the historical data itself tends to be limited in both relevance and size, due to the rapid evolution of distribution systems: network topologies are frequently reconfigured and new DERs, such as rooftop solar units, are continuously being installed. As a result, previously collected data may quickly become outdated or uninformative.

These two features, namely, structural unobservability and limited historical data, characterize the operational setting considered in this work. While several approaches to distribution system state estimation have addressed information limitations, a comprehensive treatment of the combined scenario involving structural unobservability and limited historical data remains largely unexplored in the literature, despite its growing practical relevance. In this paper, we propose estimation methods that are explicitly designed to operate under such information-constrained conditions, offering improved robustness and accuracy where conventional methods are not implementable or tend to fail.

Such a dramatic information shortage motivates the use of contextual learning approaches, which aim to exploit similarities between present and past measurements to approximate the conditional distribution of the underlying system state. However, this approximation is highly susceptible to inference errors, particularly when the available historical dataset is sparse or when the same real-time measurements may correspond to a broad range of possible system states [4].

A prevalent strategy in the literature to handle previous challenges is the adoption of Bayesian frameworks. For instance, [5] employs a deep neural network to learn mappings from limited real-time measurements to system states, combining it with a Bayesian state estimation algorithm to improve robustness against bad data. In [6], Gaussian process-

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based load forecasts are integrated with a linearized three-phase power flow model, enabling voltage predictions that are subsequently updated via linear Bayesian estimation. The authors in [7] introduce a scalable and robust Gaussian process regression technique that supports accurate pseudo-measurement inference and state estimation using a limited number of measurements.

Alternatively, some works adopt data-driven corrections to physical models. For example, [8] proposes a data-enhanced linear power flow formulation to improve the accuracy of state estimation functions, in which the elements of the admittance matrix are adjusted based on available measurements. Robustness is ensured through a Huber estimator, allowing convergence even under uncertain model parameters, while zero-injection nodes are utilized to enhance observability. In a different direction, [9] formulates the problem as a mixed-integer linear program (MILP), treating both system states and load consumptions as decision variables. The aim is to maximize state estimation accuracy under power flow constraints, with pseudo-measurements from historical data setting feasible bounds for load variables. With a similar mathematical approach, [10] addresses the challenge of topology identification in unobservable distribution systems with limited measurements. A compressive sensing framework that jointly estimates system state and network topology is proposed, formulated as a MILP with auxiliary variables to eliminate non-linearity and a convex relaxation to enable faster solution times. Finally, in [11] a modified weighted least squares (WLS) estimator is designed to operate effectively with very few real measurements and without relying on pseudo-measurements, even in typically unobservable scenarios. The method retains the advantages of a mature and widely used state estimation approach while enhancing observability by incorporating allocation factors, whose key assumption is that the power consumed (or generated) within each cluster of buses would be always at the same per-unit level (with respect to the rated one) for each node of the grid.

Despite their contributions, the described literature either assumes access to large amounts of historical data [5]–[8], relies on approximated models [8]–[10], or introduces structural simplifications that reduce the complexity of the state estimation [11]. Against this background, the contribution of this paper is threefold:

- We first introduce a non-parametric data-driven state estimator which leverages nearest neighbors to approximate the conditional expectation of the delayed measurements' residuals given the available measurements. The estimator is expressed as a contextual stochastic optimization problem.
- We then present a robustified extension of this state estimator, which optimizes against ambiguity on the conditional distribution within a vicinity of the one estimated by the nearest neighbors. This novel formulation aims to reduce sensitivity to sample uncertainty and sparsity, inherent in the presented scenario of limited real-time data. Through a set of numerical experiments, we analyze the conditions under which this added robustness translates into improved estimation performance.
- Finally, as a practically important feature of our proposal, we show that the robust contextual state estimator remains nearly as simple as the standard one from a computational standpoint, requiring only a modest increase in problem size through the introduction of auxiliary variables and constraints.

The rest of the paper is organized as follows. In Section II we present the state estimation problem, emphasizing the challenges posed by unobservable systems and detailing the proposed approximation of the conditional distribution of delayed measurements given the real-time available ones. Both the non-robust baseline and the robust estimator are then presented, highlighting their respective advantages and limitations. Section III provides numerical results that assess the performance of the proposed method against relevant benchmarks. Finally, Section IV summarizes the key findings and outlines directions for future research.

II. Methodology

This section begins by presenting the conventional static state estimation problem in power systems, outlining its key components and highlighting a critical limitation of practical relevance: the lack of real-time observability due to the sparsity of available measurements. Building on this, we introduce two data-driven methods designed to estimate the system state under partial real-time observability and limited historical data. Both methods rely on a K-nearest neighbors (K-NN) approach to approximate the conditional expectation of the residuals associated with delayed measurements, given the available real-time data. The second method (our proposal) incorporates a robustification mechanism aimed at mitigating inference errors caused by sample scarcity and the limited informativeness of the measurements. We then formalize this robust estimator mathematically and devote a separate section to detailing the data-driven procedure used to tune the robustness parameter that governs its behavior.

Using generic notation, the relationship between the measurement vector $z \in \mathbb{R}^m$ and the state of the system $x \in \mathbb{R}^n$ is characterized as follows.

$$z = h(x) + e \tag{1}$$

where $h(\cdot) : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the measurement function, which is constructed based on the power flow equations, the physical laws that govern the relationship between the measured magnitudes and the state variables. The term $e \in \mathbb{R}^m$ denotes the measurement error vector, whose components are typically assumed to follow normal distributions with zero mean and to be independent and uncorrelated with respect to the state variables [12]. Given the previous assumptions, the state estimation problem can be formulated as a WLS optimization problem [12]. This problem seeks to minimize the squared Euclidean norm of the weighted measurement errors:

$$\hat{x} \in \arg\min_{x} \|W(z - h(x))\|_2^2 \tag{2a}$$

$$\text{s.t. } h^z(x) = 0 \tag{2b}$$

where $\hat{x}$ is the estimated state vector that can be decomposed into voltage magnitude $\widehat{V}$ and angle $\hat{\theta}$, and the weight matrix $W = \text{diag}(\sigma_1^{-2}, \ldots, \sigma_m^{-2})$ represents the user's confidence in the measured data. Here, σ_m^2 represents the error variance of the m-th measurement. Equation (2b) acts as a strong physical constraint at zero-injection nodes, condition that enhances the observability of distribution systems and must be fully leveraged [8]. For ease of notation, the solution of problem (2) is denoted as $\hat{x} = \texttt{wls}(z)$. In the general case, problem (2) results in a non-convex, constrained optimization problem, which has conventionally been solved using an iterative Newton-Raphson algorithm [13]. Alternatively, problem (2) can also be directly solved using interior-point methods implemented in non-linear optimization solvers [14].

As discussed in Section I, even when the number and types of measurements captured by the sensors are sufficient to make the system fully observable, some measurements may not be available in real time due to delays or communication issues. In both this study and in prior work [15], we consider the measurement set divided into two groups: fast and sparse and slow but abundant. Consequently, we denote the set of available real-time measurements as $z^a \in \mathbb{R}^{m_a}$ and the set of delayed measurements as $z^d \in \mathbb{R}^{m_d}$. We then split the measurement function as follows:

$$z^a = h^a(x) + e^a \tag{3a}$$

$$z^d = h^d(x) + e^d \tag{3b}$$

where $h^a(\cdot) : \mathbb{R}^n \to \mathbb{R}^{m_a}$, $h^d(\cdot) : \mathbb{R}^n \to \mathbb{R}^{m_d}$, $e^a \in \mathbb{R}^{m_a}$, $e^d \in \mathbb{R}^{m_d}$, and $m = m_a + m_d$. In this work, we assume that the number and type of measurements available in real time are *not* sufficient to make the system observable [3], [16]. According to [17], the system is numerically observable if the measurement Jacobian matrix is full rank and well-conditioned.

Although model (2) cannot be used to infer the state of the system in real time in the present instance t, denoted as $\hat{x}_t$, due to the lack of observability, we can always perform a *retrospective* state estimation for any past instance s with measurements (z_s^a, z_s^d). This retrospective system state is denoted as $\hat{x}_s^d$ and can be computed as $\hat{x}_s^d \in \texttt{wls}(z_s^a, z_s^d)$. We assume that, when performing state estimation for the current instance t, we have access to the enlarged historical dataset $\{(z_s^a, z_s^d, \hat{x}_s^d),\ \forall s \in \mathcal{S}_t\}$, where $\mathcal{S}_t$ denotes the subset of historical data relevant at present time t. In line with the online nature of the setting, we further assume that $\mathcal{S}_t$ has small cardinality, consistent with the idea that only the most recent past instances are informative for estimating the system state at time t.

Although in practice we only have access to the available measurements z_t^a, we can still estimate the conditional distribution of the delayed measurements z_t^d given z_t^a using a data-driven, non-parametric approach. Specifically, the K-NN method selects the K historical samples whose available measurement vectors are closest to z_t^a, identifying the corresponding set of indices $\mathcal{S}_t^K$ as follows:

$$\mathcal{S}_t^K := \arg \min_{\substack{\mathcal{K} \subseteq \mathcal{S}_t \\ |\mathcal{K}| = K}} \sum_{k \in \mathcal{K}} \|W^a(z_t^a - z_k^a)\|_2^2. \tag{4}$$

Consequently, the conditional distribution of the delayed measurements given the available ones $p(z_t^d | z_t^a)$ is approximated by the points $\{z_s^d, \forall s \in \mathcal{S}_t^K\}$.

In the following lines, we present two data-driven methods for estimating the system state under partial real-time observability and limited historical data. The first is a standard K-NN approach, which leverages nearest neighbors to approximate the conditional expectation of the delayed measurements' residuals given the available measurements. Although effective in many cases, this method is sensitive to noise and sample sparsity, inherent challenges in power system state estimation with limited data. To address this, we propose an extension inspired by distributionally robust optimization (DRO) with contextual information that explicitly accounts for uncertainty in the conditional inference process. By optimizing against ambiguity on the conditional distribution within a neighborhood of the one estimated by the K-NN, this robust formulation reduces inference errors, particularly in scenarios where data are scarce or non-representative.

A. Vanilla K-Nearest Neighbor State Estimator

The first of the data-driven methods selects the K historical instances most similar to the real-time available measurements using (4) and searches for a state that best conforms with both the available measurements z_t^a and the conditional probability distribution of the delayed measurements given the available measurements $p(z_t^d | z_t^a)$ as estimated by the nearest neighbors.

Alternatively to the common approach of constructing pseudo-measurements [15], we adopt a *stochastic* approach using the residuals of the delayed measurements, allowing the state estimator to explicitly account for their associated uncertainty. We denote this method *vanilla* to differentiate it from the robust extension presented afterwards. The vanilla K-NN state estimate $\hat{x}_t^v$ is obtained by solving the non-linear weighted least squares problem (5), initialized with a flat voltage.

$$\hat{x}_t^v \in \arg\min_x \|W^a(z_t^a - h^a(x))\|_2^2 + \frac{1}{K} \sum_{s \in \mathcal{S}_t^K} \|W^d(z_s^d - h^d(x))\|_2^2 \tag{5a}$$

$$\text{s.t.}\ \ h^z(x) = 0 \tag{5b}$$

This method is a *lazy learning* approach, which means that no offline model training is required. Instead, the optimization problem is performed online using a local selection from the historical dataset. The procedure is described in Algorithm 1, where we denote $\hat{x}_t^r \in \texttt{Vanilla}(\mathcal{S}_t^K, z_t^a)$, emphasizing its dependence on the K-nearest neighbors and the available measurements. This approach is inherently sensitive to measurement noise and sample uncertainty, especially in the few-training-sample setting considered. To address this, we introduce a robust counterpart grounded in DRO with contextual information.

B. Contextual Robust State Estimator

The vanilla K-NN algorithm approximates the conditional distribution of the delayed measurement vector z^d given the

Algorithm 1 *Vanilla*

Input: number of neighbors, K; training set, $\{(z_s^a, z_s^d)\ \forall\, s \in \mathcal{S}_t\}$; available measurements for test instance z_t^a.

1) Compute $\mathcal{S}_t^K$ using (4).
2) Solve (5): $\hat{x}_t = \hat{x}_t^v \in \texttt{Vanilla}(\mathcal{S}_t^K, z_t^a)$.

Output: State Estimation $\hat{x}_t$.

available measurement vector z_t^a (referred to as the *context*) using a discrete uniform distribution supported on the K training samples closest to z_t^a. The underlying intuition is that training samples in close proximity to z_t^a are more likely to originate from a distribution similar to the target one. However, this approximation is prone to significant inference errors, particularly when the training dataset is small or in power systems where a single context z_t^a can correspond to a wide variety of system states. To enhance robustness against such inherent estimation errors, we draw inspiration from [18] and propose a simple yet effective *Robust Contextual State Estimator*, based on the min-max formulation (6).

In (6), the inner maximization identifies the most adversarial reweighting of the neighbors, subject to the three conditions defined by constraints (6b)–(6e). Specifically, constraint (6b) ensures that no individual neighbor is assigned a weight greater than $\frac{1}{K}$, which is equivalent to requiring that at least K neighbors receive non-zero weights. Constraint (6c) enforces that the weights form a valid probability distribution with finite discrete support. Lastly, constraint (6e) bounds the squared, probability-weighted distance between the neighbors and the context z_t^a by a user-defined hyperparameter ρ. Hence, this parameter plays the role of a tolerance margin that limits the allowable contextual deviation between the current available measurement vector z_t^a and the historical available measurements z_s^a. This ensures that the constructed probability distribution remains contextually close to z_t^a, while reducing the impact of the potential misalignment between similar observations and dissimilar underlying states. Expressed in the terminology of DRO, ρ controls the size of the ambiguity set and thereby governs the degree of reliance on the conditional distribution estimated by the K-NN.

Finally, the outer minimization searches for a state estimate x that conforms with both the available measurements and the least favorable reweighting of the neighbors under the three restrictions mentioned above. In doing so, the system state estimation is made robust against the unavoidable inference error incurred by the vanilla K-NN.

$$\hat{x}_t^r \in \arg\min_{x} \max_{w_s \geq 0} \sum_{s\in\mathcal{S}_t} w_s ||W^a(z_t^a - h^a(x))||_2^2 + \sum_{s\in\mathcal{S}_t} w_s \|W^d(z_s^d - h^d(x))\|_2^2 \tag{6a}$$

$$\text{s.t. } w_s \leq \frac{1}{K}, \quad \forall\, s \in \mathcal{S}_t \quad : \lambda_s \tag{6b}$$

$$\sum_{s\in\mathcal{S}_t} w_s = 1 \quad : \mu_1 \tag{6c}$$

$$\sum_{s\in\mathcal{S}_t} w_s ||W^a(z_t^a - z_s^a)||_2^2 \leq \rho \quad : \mu_2 \tag{6d}$$

$$h^z(x) = 0 \tag{6e}$$

Unlike the general distributionally robust approach with contextual information introduced in [18], model (6) possesses a distinct feature that makes it particularly appealing for enhancing robustness in the non-convex optimization involved in state estimation: it can be efficiently reformulated as the equivalent single-level minimization problem (7), which remains *as tractable as* the vanilla K-NN state estimator (5). This reformulation requires only the addition of $|S_t| + 2$ variables and $2|S_t| + 1$ constraints (recall that the size of S_t is small) and leverages the fact that the inner maximization in (6) is a linear program. As such, the transformation is achieved by deriving the dual of the inner maximization problem (with dual variables corresponding to each set of constraints in (6) indicated after a colon) and combining the resulting minimization with the original outer problem as follows:

$$\hat{x}_t^r \in \arg\min_{x,\, \lambda_s,\, \mu_1,\, \mu_2} \frac{1}{K} \sum_{s\in\mathcal{S}_t} \lambda_s + \mu_1 + \rho\, \mu_2 \tag{7a}$$

$$\text{s.t. } \lambda_s + \mu_1 + \mu_2 \|W^a(z_t^a - z_s^a)\|_2^2 \geq \|W^a(z_t^a - h^a(x))\|_2^2 + \|W^d(z_s^d - h^d(x))\|_2^2, \quad \forall s \in \mathcal{S}_t \tag{7b}$$

$$\mu_2 \geq 0 \tag{7c}$$

$$\lambda_s \geq 0, \quad \forall s \in \mathcal{S}_t \tag{7d}$$

$$h^z(x) = 0 \tag{7e}$$

For a given value of the robustness parameter ρ, we solve problem (7) in two steps, where the first one is intended to identify a good initial feasible solution. For this purpose, we warm-start the optimization by fixing the state estimate x to the value obtained from the vanilla estimator, denoted $\hat{x}_t^v$. This choice is motivated by the fact that, by solving (7), we seek to improve on the vanilla K-NN estimate. With x fixed, the problem becomes linear in the dual variables λ_s, μ_1, and μ_2. We solve this simplified problem to obtain initial estimates $\hat{\lambda}_s$, $\hat{\mu}_1$, and $\hat{\mu}_2$ for the dual variables.

In the second step, we use these dual variable estimates together with the associated vanilla state estimate $\hat{x}_t^v$ to initialize and solve the full non-linear optimization problem (7) over both x and the dual variables. This second step thus delivers the target contextual robust state estimate. Although the full problem is non-convex due to the non-linear dependence of the measurements on the state (via $h^a(x)$ and $h^d(x)$), this two-stage approach helps guide the solver towards a better local optimum and accelerates convergence. We denote the final robust estimate resulting from this procedure as $\hat{x}_t^r \in \texttt{Robust}(\mathcal{S}_t, z_t^a, K, \hat{x}_t^v, \rho)$, emphasizing its dependence on the training set, the available measurements, the number of nearest neighbors, the initialization from the vanilla estimate, and the chosen robustness level.

C. Selection of the Robustness Parameter ρ

The regularization parameter ρ in (7) captures how informative the available measurements are about the delayed ones by limiting the contextual mismatch between the current vector z_t^a and the historical samples z_s^a, ensuring that only relevant past instances contribute to the estimation. Selecting an appropriate value of ρ is therefore critical to achieving a good trade-off between robustness to inference errors and estimation accuracy. To define a sensible data-driven range for the robustness parameter ρ, we introduce two extreme values based on the structure of the problem. Since constraint (6b) enforces that at least K neighbors must receive non-zero weights, the feasibility of the problem requires that ρ be no smaller than the average contextual distance of the K-nearest neighbors. This value thus defines a natural lower bound. Conversely, a loose upper bound for ρ can be set as the contextual distance of the furthest neighbor in the dataset. These bounds are formalized in (8b)–(8c). Consequently, we use a geometric progression to divide the search range into logarithmically spaced values. The resulting sequence $\{\rho_c \ \forall c \in \mathcal{C}\}$ is computed using (8), where $\mathcal{C}$ is the set of natural numbers from 1 to $|\mathcal{C}|$.

$$\rho_c = \exp\left(\log(\rho_{\min}) + \frac{c-1}{|\mathcal{C}|-1}\left(\log(\rho_{\max}) - \log(\rho_{\min})\right)\right) \tag{8a}$$

$$\rho_{\min} = \frac{1}{K}\sum_{s\in\mathcal{S}_t^K} \|W^a(z_t^a - z_s^a)\|_2^2 \tag{8b}$$

$$\rho_{\max} = \max_{s\in\mathcal{S}_t} \|W^a(z_t^a - z_s^a)\|_2^2 \tag{8c}$$

Once defined the search range, we first perform an online grid search over the set of candidate values $\{\rho_c \ \forall c \in \mathcal{C}\}$ in order to obtain the corresponding states $\hat{x}_{t,c}^r$. Then, for each candidate ρ_c, we calculate the average Euclidean distance between the resulting state $\hat{x}_{t,c}^r$ and the retrospective states of the K-nearest neighbors in the historical samples using (9).

$$E_{t,c} = \frac{1}{K}\sum_{s\in\mathcal{S}_t^K} ||\hat{x}_{t,c}^r - \hat{x}_s^d||_2^2 \tag{9}$$

The index c^* that minimizes the estimation error validated across the K-NN is selected using (10) for the final estimation. This complete procedure, denoted RC-SE (from *Robust Contextual State Estimator*), is presented in Algorithm 2.

$$c^* = \arg\min_{c\in\mathcal{C}} E_{t,c}, \quad \hat{x}_t = \hat{x}_{t,c^*}^r \tag{10}$$

Algorithm 2 RC-SE

Input: number of neighbors, K; measurements and retrospective state in training set, $\{(z_s^a, z_s^d, \hat{x}_s^d) \ \forall s \in \mathcal{S}_t\}$; available measurements and vanilla state estimate for test instance, $\{z_t^a, \hat{x}_t^v\}$; set of robust parameter values, $\{\rho_c \ \forall c \in \mathcal{C}\}$.

1) Solve (7) : $\hat{x}_{t,c}^r = \texttt{Robust}\,(\mathcal{S}_t, z_t^a, K, \hat{x}_t^v, \rho_c)\,\forall c \in \mathcal{C}$.
2) Compute $E_{t,c} = \frac{1}{K}\sum_{s\in\mathcal{S}_t^K} ||\hat{x}_{t,c}^r - \hat{x}_s^d||_2^2 \,\forall c \in \mathcal{C}$
3) Compute $c^* = \arg\min_{c\in\mathcal{C}} E_{t,c}$
4) Select $\hat{x}_t = \hat{x}_{t,c^*}^r$

Output: State Estimation $\hat{x}_t$

III. Case study

In the following, we present results from a series of numerical experiments designed to evaluate the performance of our proposed methodology under varying operating conditions. We begin by outlining the experimental setup, which systematically explores the influence of four key elements: network complexity, load variability, training set size, and measurement configuration. This is followed by breaking down each of the estimators proposed in Section II and comparing them against relevant benchmarks, detailing the metrics used. In the final part of the section, we discuss the observed results in depth, highlighting how each factor impacts performance and illustrating the consistency and benefits of the robust formulation across all scenarios.

A. Setup

The effectiveness of the methodology presented in Section II is highly sensitive to the nature of the measurement functions h, that is, how variations in the available measurements z_t^a relate into changes in the true underlying state x_t. This mapping governs the quality of both the nearest-neighbor estimation capabilities and the benefits of increasing robustness for the state estimation. In this case study, we systematically analyze the key factors that influence this relationship:

I. the degree of complexity of the underlying system,
II. the variability of the operating conditions,
III. the size of the training dataset, and
IV. the configuration of the measurement set.

By isolating and varying each of these elements, we aim to understand their impact on estimation accuracy and the relative performance of the proposed data-driven methods. In the following, we discuss the effect of each element.

1) Complexity of the network: The extent to which the uncertainty arising from delayed measurements impacts performance depends heavily on the complexity of the underlying power system. To explore this, we consider two contrasting case studies. The first is the IEEE 39-bus system, which features a meshed network topology, multiple generators distributed across the grid, and operating conditions that induce strong non-linear relationships between variables. In such a system, similar values of available measurements can correspond to significantly different delayed measurements. As a result, the use of a robust estimation strategy becomes particularly valuable, and clear improvements over the standard K-NN approach are expected. By contrast, the IEEE 33-bus system represents a simpler scenario. It has a radial structure, a single generating unit, and more linear measurement functions (as reported in [19]), which lead to a more predictable relationship between variables. In this case, the uncertainty in the distribution estimated from data is less impactful, and the advantages of the robust methodology, while still present, are expected to be more modest. All system data are sourced from the test case library of the open-source tool Pandapower [20].

2) Variability of operating conditions: To train and evaluate the learning-based estimators described in Section II, we generate diverse and realistic datasets using Pandapower by performing Optimal Power Flow (OPF) simulations with synthetic load profiles, created by stochastically varying both active and reactive power demands. We generate $10\,000$ instances for the simple IEEE 33-bus network due to its smooth state-measurement mapping, while we opt for $100\,000$ instances for the IEEE 39-bus to capture its higher complexity. For each instance, the active power demand is distributed across buses using normalized load participation factors derived from the original system demand. Specifically, the share of total system demand assigned to each load bus is randomly perturbed within a ±δ% range with $\delta = \{1, 5, 20\}$, and then renormalized to preserve the total system demand. The reactive power demand is determined by varying the power factor of each load around its nominal value. This variation is also randomized within a specified interval following the uniform distribution $\mathcal{U}(0.7\cos\phi, 1.3\cos\phi)$ for the simple network and $\mathcal{U}(0.975\cos\phi, 1.025\cos\phi)$ for the complex, to ensure OPF convergence, while guaranteeing the power factor does not exceed 1.

3) Training size: From the previously defined dataset, we randomly selected 400 instances as test samples. Our focus is on evaluating the proposed methodology's ability to generalize from limited but contextually similar data, a scenario commonly encountered in online tasks such as state estimation, where recently solved instances can be highly informative for solving current ones. To simulate this, we restrict the training set to the $|\mathcal{S}_t|$ nearest neighbors with respect to the available measurements z_t^a of each test instance, using (4). In order to study the impact of training size, we select between $|\mathcal{S}_t| = \{20, 50, 100\}$ and since we consider $K \approx \sqrt{|\mathcal{S}_t|}$, we correspondingly choose $K = \{5, 7, 10\}$.

4) Measurement set: The final factor that directly shapes the relationship between the available measurements z_t^a and the system state x is the number and type of available measurements. Since the two networks under study differ slightly in size and topology, we ensure a fair comparison by maintaining comparable measurement redundancy, quantified as $R = m^a/n$, where m^a is the number of real-time available measurements and n is the number of state variables in x. As defined in Section I and in [2], we classify PMUs (voltage magnitude and angle) and SCADA measurements (active and reactive power injections) as real-time available, while SM (active and reactive power injections in the remaining nodes) are considered delayed due to slower reporting times [3]. The system measurements are obtained by adding a Gaussian noise with standard deviation $\sigma_{P,Q} = 0.01$ to power injections and flows and $\sigma_{V,\theta} = 0.001$ to voltage magnitudes and angles, a similar process as in [21]. In order to analyze the effect of different measurement setups, we design four scenarios with increasing redundancy, collected in Table I.

Note that in all scenarios, the system remains unobservable in real time, meaning that its state cannot be reconstructed using conventional estimation techniques based solely on the available measurements. However, for the purpose of training and validation, we assume that both the real-time available and delayed measurement sets are fully accessible retrospectively.

TABLE I: Measurement setup configurations by redundancy level.

Redundancy	R	SCADA	PMU
Lowest	0.40	Low	No
Low	0.55	Low	Yes
High	0.70	High	No
Highest	0.90	High	Yes

B. Performance comparison

In this subsection, we introduce the learning-based estimators considered in our study and define the metrics used for performance comparison. Thus, we define the following approaches:

- *Vanilla*: This method corresponds to the vanilla K-NN state estimator described in Section II-A and in Algorithm 1. It solves a weighted least squares problem using the K closest historical delayed measurements z_s^d according to the available measurements z_t^a. No robustness or ambiguity modeling is applied in this method.
- RC-SE: This method is presented in Section II-B and corresponds to the contextual robust extension of the *Vanilla*. It leverages available measurements to estimate state variables in a manner that is robust to the uncertainty introduced by delayed measurements, which must be inferred from historical samples. This uncertainty arises from the variability in the relationship between available and delayed measurements, a challenge that becomes more pronounced when historical data is scarce. The full procedure is detailed in Algorithm 2.
- *Retrospective*: This baseline assumes that all measurements are available and the system is then observable, that is, $\hat{x}_t = \hat{x}_t^d$. Under the considered setup, however, this situation is unrealistic and only used for benchmarking purposes.
- *Anticipative*: This baseline simulates an oracle version of the RC-SE, in which the retrospective state $\hat{x}_t^d$ of the test instance is assumed to be known and available for hyperparameter tuning. While this benchmark is not implementable in practice, it enables us to assess how the availability of high-quality information for tuning the parameter ρ during validation influences overall performance.
- *Persistent*: This baseline uses the state of the closest neighbor relative to the available measurements z_t^a for each test instance, that is, $\hat{x}_t = \hat{x}_s^d$, $s \in \mathcal{S}_t^1$. It avoids solving any optimization problem, thus the computation is negligible. This method serves as a naive but relevant benchmark in the presented scenario of contextually similar training data.

In order to compare the performance of the previous approaches, we use the Root Mean Squared Error (RMSE) defined as:

$$\text{RMSE}_x = \sqrt{\frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} ||\hat{x}_t - \tilde{x}_t||_2^2} \tag{11}$$

where $\mathcal{T}$ is the set of instances in the test dataset, $\hat{x}_t$ is the estimated state by each approach and $\tilde{x}_t$ is the ground-truth state, obtained from OPF simulations as described in Section III-A. Alternatively, we also compute the accuracy metric (11) for the derived power variables, with respective p.u. values of apparent power injections and apparent power flows, correspondingly denoted RMSE_S. Finally, we select $|\mathcal{C}| = 20$ in (8) and the state estimation problems (5) and (7) are modeled in Pyomo 6.8.0 [22] running in Python 3.11.10 and solved with Knitro 14.0.0 [23]. The data generated for this case study can be downloaded from [24].

C. Results and discussion

To facilitate a detailed comparison of the methods introduced in Section III-B, we begin by analyzing a representative scenario selected from the range of configurations discussed earlier. Specifically, Table II reports estimation accuracy, measured as defined in (11), under the high measurement redundancy scenario (as presented in Table I) and fixed conditions of moderate load variability and training set size ($\delta = 5$, $|\mathcal{S}_t| = 50$, and $K = 7$), for both networks. The best-performing method in each category is highlighted in bold.

First, the results confirm that state estimation in the more complex IEEE 39-bus system is significantly more challenging than in the simpler IEEE 33-bus network. Nonetheless, in both cases, the proposed RC-SE consistently outperforms the *Vanilla* estimator, validating the benefits of accounting for distributional uncertainty. While the *Persistent* method yields slightly better state estimates than *Vanilla*, it performs substantially worse on derived power quantities, highlighting the need to assess estimator performance across all relevant variables when real-time measurements are limited.

Importantly, the RC-SE estimator closely tracks the performance of the idealized *Anticipative* method, suggesting that the validation procedure outlined in Section II-C is effective for selecting the robustness parameter ρ. Table II also illustrates that the gains from incorporating distributional robustness are far more pronounced in the IEEE 39-bus system, with relative improvements exceeding 52.7% for state variables and 20.0% for power quantities, compared to the *Vanilla* approach. For the IEEE 33-bus system, these improvements are more modest, at 4.6% and 2.4%, respectively.

An especially noteworthy finding is that the non-implementable *Retrospective* method performs worst on the IEEE 33-bus system, despite outperforming all other methods in the 39-bus case. This counterintuitive result highlights the high predictability of the 33-bus system's state from historical data, which stems from the system operating within a regime where the measurement function exhibits approximately linear behavior, along with its simple radial topology and network configuration. These characteristics are effectively exploited by the KNN-based estimators, particularly those that minimize the conditional expectation of delayed measurement residuals. This finding suggests that, in traditional distribution systems, the operational patterns encoded in past data can be as informative as real-time physical measurements for state estimation. Lastly, recall that the robustness parameter in *Anticipative* is overfitted to the *Retrospective* estimator, and thus, the performance of the former is inherently tied to that of the latter.

TABLE II: Test set RMSE values ($\times 10^{-3}$) for all methods and both networks, in the high measurement redundancy scenario with $\delta = 5$, $|\mathcal{S}_t| = 50$ and $K = 7$.

Network	Method	RMSE_x	RMSE_S
33-bus (simple)	Vanilla	2.59	9.28
	RC-SE	**2.47**	**9.05**
	Persistent	2.88	10.30
	Anticipative	2.56	9.12
	Retrospective	2.86	10.50
39-bus (complex)	Vanilla	7.14	68.31
	RC-SE	3.38	54.68
	Persistent	4.28	85.15
	Anticipative	3.22	52.25
	Retrospective	**1.03**	**9.26**

TABLE III: RC-SE vs. *Vanilla* test set RMSE ($\times 10^{-3}$) values for both networks, in the high measurement redundancy scenario with $\delta = 5$.

Network	$\lvert\mathcal{S}_t\rvert$	K	RMSE_x		RMSE_S	
			Vanilla	RC-SE	Vanilla	RC-SE
33-bus (simple)	20	5	2.60	2.50	9.44	9.14
	50	7	2.59	2.47	9.28	9.05
	100	10	2.55	2.47	9.17	9.00
39-bus (complex)	20	5	7.50	3.42	70.77	55.61
	50	7	7.14	3.38	68.31	54.68
	100	10	6.92	3.39	66.46	54.33

In what follows, we focus our analysis on comparing the *Vanilla* and RC-SE methods. The influence of the training set size across network types is assessed in Table III. The reported test set RMSE values exhibit only modest variations as the training set grows, with a slight improvement generally observed. This behavior is expected, given that both training and test samples in this experiment were generated under a load variability of only 5%. Under these conditions, both the *Vanilla* and the proposed RC-SE estimators benefit from a larger and more informative set of K-nearest neighbors. In stark contrast, we can see in Table IV that, when the load variability increases (mimicking systems with faster dynamics and, consequently, delivering less informative neighbors), the performance of both methods deteriorates for the complex network. This is unsurprising because our approach is tailored for online environments where recent measurements are assumed to have been generated from similar system states and thus hold significant predictive power for the next state. In any case, the proposed RC-SE consistently and significantly outperforms the *Vanilla* estimator across all scenarios. It is worth noting, however, that the relative improvement tends to decrease as load variability grows, due to the diminishing statistical representativeness of the training set under more dynamic conditions.

The final factor to be assessed is the measurement set itself. To evaluate its impact on performance, Table V presents test set RMSE values under fixed conditions of moderate load variability and training size ($\delta = 5$, $|\mathcal{S}_t| = 50$, and $K = 7$). The results confirm that RC-SE consistently outperforms the *Vanilla* estimator across both the IEEE 33-bus (simple) and 39-bus (complex) networks, for all four measurement configurations

TABLE IV: RC-SE vs. *Vanilla* test set RMSE ($\times 10^{-3}$) values for both networks across different load variability levels in the high measurement redundancy scenario with $|\mathcal{S}_t| = 50$ and $K = 7$.

Network	Variability	RMSE$_x$		RMSE$_S$	
		Vanilla	RC-SE	Vanilla	RC-SE
33-bus (simple)	±1%	2.58	2.47	9.27	9.05
	±5%	2.59	2.47	9.28	9.05
	±20%	2.51	2.41	9.03	8.84
39-bus (complex)	±1%	6.61	2.10	43.02	24.20
	±5%	7.14	3.38	68.31	54.68
	±20%	11.78	8.41	220.79	185.83

TABLE V: RC-SE vs. *Vanilla* test set RMSE ($\times 10^{-3}$) values for both networks, across different measurement redundancy levels with $\delta = 5$, $|\mathcal{S}_t| = 50$ and $K = 7$.

Network	SCADA	PMU	RMSE$_x$		RMSE$_S$	
			Vanilla	RC-SE	Vanilla	RC-SE
33-bus (simple)	Low	No	2.75	2.59	8.75	8.47
	Low	Yes	1.11	0.96	7.94	6.42
	High	No	2.59	2.47	9.28	9.05
	High	Yes	1.15	1.08	8.22	7.88
39-bus (complex)	Low	No	8.97	4.37	93.37	77.36
	Low	Yes	4.19	2.71	84.78	51.39
	High	No	7.14	3.38	68.31	54.68
	High	Yes	3.17	2.45	61.92	44.27

corresponding to the increasing levels of redundancy described in Table I. As in previous experiments, the performance gains are more pronounced in the more complex network. Remarkably, for the IEEE 39-bus system, the proposed RC-SE not only halves (or nearly halves) the state estimation error relative to the *Vanilla* estimator in all cases, but also achieves a relative improvement comparable to that obtained by incorporating high-quality PMU measurements into the available set. This highlights the substantial value of distributional robustness in complex operational contexts.

We conclude this section by noting that the results in Table V also reveal that the inclusion of SCADA and PMU measurements in the available set systematically improves the estimation accuracy of both the *Vanilla* and RC-SE estimators, with the latter consistently achieving superior performance. Interestingly, in the highly predictable 33-bus system, the simultaneous inclusion of SCADA and PMU measurements is not synergistic; in fact, it appears to slightly degrade performance. This suggests that, for this network, adding SCADA data in the presence of high-quality PMU measurements may hinder the identification of better neighbors, likely due to unnecessary redundancy in the feature space.

TABLE VI: RC-SE vs. *Vanilla* for both networks: Improvement across different measurement redundancy levels with $\delta = 5$, $|\mathcal{S}_t| = 50$ and $K = 7$.

Network	Redundancy	SCADA	PMU	$\Delta_x(\%)$	$\Delta_S(\%)$
33-bus (simple)	Lowest	Low	No	5.81	3.22
	Low	Low	Yes	**13.70**	**19.09**
	High	High	No	4.58	2.39
	Highest	High	Yes	6.77	4.03
39-bus (complex)	Lowest	Low	No	51.22	17.16
	Low	Low	Yes	35.14	**39.39**
	High	High	No	**52.68**	19.95
	Highest	High	Yes	22.76	28.51

IV. Conclusion

This work addresses the critical challenge of state estimation in distribution systems under a double information shortage: on the one hand, structural unobservability, where the set of real-time available measurements does not suffice to determine the system state; and on the other hand, limited access to historical data, due to the evolving nature of distribution networks, where frequent topology reconfigurations and continual integration of new DERs reduce the usefulness and representativeness of past data. These two features characterize the operational setting considered in this work, and motivate the development of estimation methods specifically designed for such conditions.

The proposed methodology combines a non-parametric, nearest-neighbor-based estimator with a robustification step that mitigates the effects of data scarcity and sample uncertainty. This robust estimator optimizes over a neighborhood of the empirical distribution defined by the nearest neighbors, reducing the sensitivity to potentially misleading historical samples.

The effectiveness of the approach has been assessed on two contrasting test systems, namely, the IEEE 33-bus and IEEE 39-bus networks, under a variety of scenarios involving different levels of load variability, different training sizes and different configurations of the available measurement set. These experiments confirm the relevance of our assumptions and the practical value of our method. In particular, the IEEE 39-bus system exemplifies a complex setting, with a meshed topology, multiple generators, and strong non-linear relationships between variables. In such a case, similar real-time measurements can correspond to markedly different system states, and the robust estimator provides significant relative improvements in RMSE, ranging from 15.7% to 69.7% over the standard K-NN baseline. Conversely, the IEEE 33-bus system, with its radial topology and simpler operating conditions, exhibits more predictable relationships between variables. Here, the benefits of robustification are more modest, with relative improvements in the range of 1.9% to 19.1%, depending on the estimated magnitude and load variability.

As future work, our methodology could be extended in two complementary directions. First, it is compatible with approaches aimed at estimating the parameters of the distribution network, such as those in [8], which directly impact the measurement function h and therefore introduce additional complexity to the estimation task. Second, although we have assumed throughout that the network topology (and hence h) is known, an important research avenue involves combining our robust estimator with topology identification methods, or introducing an additional robustification layer to account for uncertainty in the measurement function itself.

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